

Genetic Risk for Alzheimer Disease, Midlife Hypertension, and Dementia

The ARIC Neurocognitive Study

Valerie N. Morrill,¹ James Russell Pike,² Jiaqi Hu,³ Myriam Fornage,⁴ Aditya Surapaneni,⁵ Keenan A. Walker,⁶ David S. Knopman,⁷ Thomas H. Mosley,⁸ Josef Coresh,² Andrea Lauren Christman Schneider,⁹ Jason R. Smith,¹⁰ and Rebecca F. Gottesman¹

Correspondence

Ms. Morrill
valerie.morrill@nih.gov

Neurology® 2026;107:e218280. doi:10.1212/WNL.00000000000218280

Abstract

Background and Objectives

Genetics represent a nonmodifiable risk factor for Alzheimer disease (AD), with 60%–80% heritability. Midlife hypertension is a modifiable risk factor for both dementia and death. Our primary objective was to determine how genetic risk for AD modifies the association between hypertension and dementia.

Methods

The Atherosclerosis Risk in Communities Study is an ongoing community-based prospective cohort study of 4 US centers. We analyzed White and Black participants free of dementia at age 55 years with genotypes and blood pressure measured at visit 1 (1987–1989). Three genetic risk groups (low, medium, high) were defined based on tertiles of a race-specific AD polygenic risk score. Dementia was ascertained through cognitive testing, informant interviews, hospitalization, codes and death records. Death was ascertained through the National Death Index. We examined the association of midlife hypertension with incident dementia within 3 genetic risk groups using Cox proportional-hazards and cumulative incidence function estimations. We used age 55 years as the time origin, with left truncation to allow entry at ages older than 55 years; age on December 31, 2022, was the administrative censoring date.

Results

Among 8,931 White and 2,666 Black participants, the median follow up time was 26.6 and 23.8 years, the mean age was 54.0/53.5 years, and 53.0%/62.5% were female, respectively. After adjusting for demographics, midlife hypertension was significantly associated with dementia incidence across all genetic risk groups among White participants (low risk hazard ratio [HR] 1.29; 95% CI 1.07–1.55, medium risk HR 1.34; 95% CI 1.13–1.58, high risk HR 1.19; 95% CI 1.03–1.38) and among Black participants at high genetic risk (HR 1.31; 95% CI 1.04–1.66). Associations for low and medium genetic risk Black participants were consistent but not statistically significant. There were no significant differences in association of hypertension with dementia by AD genetic risk group. Individuals with hypertension had a 0%–2% higher probability of developing dementia by age 80 and a 6%–13% lower probability of dementia-free survival to age 80 years vs those without hypertension, across race and genetic risk groups.

Discussion

Genetic risk for AD does not modify the association between hypertension and dementia. These data support the fact that all individuals with hypertension are likely to benefit from antihypertensive treatment.

MORE ONLINE

Supplementary Material

¹Intramural Research Program, National Institute of Neurological Disorders and Stroke, Bethesda, MD; ²Optimal Aging Institute, New York University Grossman School of Medicine, New York; ³School of Healthcare Management, Tsinghua University, China; ⁴Health Science Center, University of Texas, Houston; ⁵Division of Precision Medicine, New York University Grossman School of Medicine, New York; ⁶Intramural Research Program, National Institute on Aging, Baltimore, MD; ⁷Mayo Clinic, Rochester, MN; ⁸University of Mississippi Medical Center, Jackson; ⁹Department of Neurology and Department of Biostatistics, Epidemiology, and Informatics, University of Pennsylvania-Perelman School of Medicine, Philadelphia; ¹⁰University of North Carolina at Chapel Hill.

Written work prepared by employees of the Federal Government as part of their official duties is, under the U.S. Copyright Act, a “work of the United States Government” for which copyright protection under Title 17 of the United States Code is not available. As such, copyright does not extend to the contributions of employees of the Federal Government.

Glossary

AD = Alzheimer disease; **ARIC** = Atherosclerosis Risk in Communities Study; **CIF** = cumulative incidence function; **DBP** = diastolic blood pressure; **GWAS** = genome-wide association study; **HR** = hazard ratio; **PRS** = polygenic risk score; **QC** = quality control; **SBP** = systolic blood pressure.

Introduction

Dementia is a progressive neurodegenerative disorder characterized by cognitive decline and memory loss, impacting 1 in 10 adults in the United States age 65 years and older.¹ The Lancet Commission identified hypertension as one of 14 modifiable targets for dementia prevention.² Hypertension can lead to significant changes in brain vasculature, white matter abnormalities, and cerebral autoregulation which can contribute to dementia.³ Reducing hypertension to improve dementia outcomes may be effective because it is highly prevalent, affecting nearly half of US adults by midlife.^{3,4} Furthermore, large epidemiologic studies show that hypertension in midlife, but not late-life, is associated with dementia^{5,6} and cognitive decline.⁷ These results suggest that hypertension in midlife leads to more severe cumulative cerebrovascular damage than hypertension in late life, or exposure to hypertension in midlife occurs at a key critical window before the development of neurodegenerative changes such as from Alzheimer disease (AD), making it an appealing target for intervention.

AD genetics are a well-established nonmodifiable risk factor for dementia. Twin studies have estimated the heritability of AD, the most common cause of dementia, to be between 60% and 80%.⁸ Carrier status of the *APOE*ε4 allele is established as a significant contributor to AD.⁹ More recently, large genome-wide association studies (GWASs) have described 85 independent risk loci in a European population¹⁰ and 5 independent loci in an African ancestry population¹¹ that are associated with AD. AD polygenic risk scores (PRSs), continuous scores that capture an individual's genetic predisposition to AD, have been used to stratify individuals into genetic risk categories.^{10,12,13} Higher AD PRS strata are associated with dementia in both European and African ancestry populations.^{12,14–16}

Characterizing the relationship between hypertension and genetic risk of AD is of interest to understand dementia pathologies and to inform targeted prevention strategies. For example, if genetic and vascular components interact, it could point to shared mechanisms and shed light on underlying etiologies of dementia. In addition, identifying AD genetic risk groups who are particularly vulnerable to the effects of midlife vascular risk factors could further support aggressive vascular risk factor management for individuals in those groups.

Using data from a large diverse population-based cohort with over 3 decades of cognitive follow-up, this study aims to assess

if the effect of midlife hypertension on dementia incidence is modified by underlying genetic risk for AD, based on published AD PRS.^{10,11} Effectively communicating the impact of modifiable risk factors across AD genetic risk strata could inform personalized prevention strategies.

Methods

Participants

The Atherosclerosis Risk in Communities (ARIC) study is an ongoing, community-based prospective cohort study. Between 1987 and 1989, ARIC enrolled 15,792 participants in midlife (44–66 years) from 4 US communities: Washington County Maryland; suburbs of Minneapolis, MN; Forsyth County, NC; and Jackson, MS.¹⁷ Participants were evaluated at enrollment (visit 1), 8 additional in-person visits through 2022, annual (semiannual since 2012) phone calls, and through ongoing surveillance. This study includes Black and White participants free of dementia at age 55 years with hypertension, genotypes, and all sociodemographic covariates (described below) measured. We also excluded participants with a suspected blood pressure measurement error (diastolic blood pressure [DBP] <30), or genotypes that did not pass quality control (QC) (eFigure 1).

Hypertension

Hypertension was defined using measured blood pressure or self-reported blood pressure medication use ascertained at visit 1. Sitting DBP and systolic blood pressure (SBP) were assessed using a random zero sphygmomanometer. After 3 consecutive measurements with >30 seconds between each measurement, SBP and DBP were defined as the mean of the last 2 measurements. Hypertension was defined as SBP ≥140 mm Hg, DBP ≥90 mm Hg, or use of blood pressure medication in the past 2 weeks.

Genotyping, AD PRS, and AD Genetic Risk Groups

Details on genotyping, QC and imputation were reported previously¹⁸ and in the eMethods. In brief, 3,207 self-identified Black and 9,743 self-identified White participants were genotyped with the Affymetrix 6.0 array. After genotype QC, sample QC, and imputation were performed separately by race group, 2,874 Black and 9,119 White participants remained. Race was self-reported and used as a proxy for genetic ancestry to apply ancestry specific PRS. However, we recognize that race is a social construct and does not fully capture underlying genetic ancestry.

We calculated a PRS for White ARIC participants based on the AD GWAS by Bellenguez et al.,¹⁰ which included 111,326 clinically diagnosed/“proxy” AD cases and 677,663 controls of European ancestry (eMethods; eTable 1). Using the 85 identified variant locations and corresponding effect sizes as weights, we calculated a PRS in White ARIC participants by calculating a weighted sum of the number of risk-increasing alleles carried for each variant using Plink2.0.¹⁹ PRS for White ARIC participants were then standardized by centering to the mean and dividing by the standard deviation. Finally, we assigned White participants to an AD genetic risk group (low risk, medium risk, or high risk) by PRS tertile.

Using the same approach, we calculated a PRS for Black ARIC participants based on the AD GWAS by Sherva et al.,¹¹ which included 13,437 clinically diagnosed/“proxy” AD cases and 69,627 controls of African ancestry (eMethods, eTable 2). For the Black ARIC participants, we used these 5 variant locations and effect sizes as weights to calculate a PRS, standardized the PRS, and assigned Black participants to an AD genetic risk group using the approach described above.

Dementia

All-cause dementia status and date of onset were assessed from visit 1 through December 31, 2022 (except through 2020 for the Jackson site), using a hierarchical approach that has been reported previously.^{20,21}

In brief, participants were tested with a standardized neuropsychological battery, consisting of 3 tests at in-person visits 2 and 4, and a longer battery covering multiple cognitive domains at visits 5 through 9. Dementia cases were identified through expert review of neuropsychological tests and informant interviews performed at visits 5–9. For individuals unable to attend in-person visits, dementia status was determined through telephone-based cognitive assessments, informant interviews, and annual/semiannual follow-up telephone questionnaires and surveillance data documenting dementia-related hospitalizations and deaths, with dementia onset dates over the entire follow-up period starting after visit 1. Event dates were estimated using this comprehensive approach, with adjustments made for individuals unable to attend in-person visits or diagnosed through surveillance.²² Individuals without dementia were censored at their last known date of follow-up or using an administrative censoring date of December 31, 2022 (except through December 31, 2020, for Jackson participants).

Death

Death was ascertained through linkage to the National Death Index.

Covariates

In our primary models, model 1 covariates included socio-demographic covariates reported at visit 1 including sex (male, female), decade of birth (1920–29, 1930–39, 1940+), and education (less than high school, high school or general

educational development or vocational school, greater than high school).

Model 2 covariates also included other potential confounders measured at visit 1, namely, body mass index (kg/m^2), total cholesterol (mg/dL), current smoker (self-reported yes/no), and diabetes (yes/no, defined as fasting glucose ≥ 126 mg/dL , nonfasting glucose ≥ 200 mg/dL , self-reported physician’s diagnosis, or diabetes medication use).

Statistical Analysis

Because we calculated ancestry-specific PRS, we conducted analyses separately among participants who identified as Black and White.²³ Participant characteristics by hypertension status were evaluated with descriptive statistics including χ^2 and *t* tests. The incidence rate of dementia per 1,000 person-years was calculated within each of the 12 AD genetic risk (low-medium-high)/hypertension (present-absent)/race (Black-White) groups. The 95% CIs were calculated using the quadratic approximation to the Poisson log likelihood for the log-rate parameter.

In the primary analysis, we used Cox proportional hazards models with age 55 years as the time origin, age as the time scale, allowing for late entries, and adjusting for model 1 covariates. To examine the association between AD genetic risk group (low-medium-high) and dementia incidence, we used 2 adjusted Cox proportional hazards models, in White and Black participants separately, with low genetic risk as the reference group. To confirm that genetic risk group provides additional information beyond *APOE*ε4 status, we repeated this analysis with AD genetic risk group membership defined using the PRS with variants in the *APOE* region removed.

Next, we used 6 adjusted Cox proportional hazards models to examine the association between hypertension and dementia, stratified by AD genetic risk group and race. Within each race group, we used an additional adjusted Cox proportional hazards model to formally test for multiplicative interaction between AD genetic risk and hypertension on risk of dementia.

To examine risk of dementia across the life course, while accounting for the competing risk of death and adjusting for mean centered model 1 covariates, we used cause-specific Cox proportional hazards models and a series of post estimations. First, within each of the 12 AD genetic risk/hypertension/race groups, we used 2 cause-specific Cox proportional hazards models. One model evaluated the association between hypertension and dementia (with death treated as censoring), and the other model evaluated the association between hypertension and death (with dementia treated as censoring). Next, we used the estimates produced by these models to derive dementia-specific and death-specific cumulative incidence function (CIF) estimations by age 70, 80, and 90, as described by Hu et al.,²⁴ and in the eMethods. The

proportional hazards assumption was confirmed by visual inspection of Schoenfeld residuals in all models.

Sensitivity Analyses

To assess for potential biases, we performed 5 sensitivity analyses. First, we examined the association of measured high blood pressure only (SBP ≥ 140 or DBP ≥ 90) and dementia incidence across genetic risk groups. Second, because participants may change blood pressure group frequently, we examined the association of hypertension at any of the first 4 study visits and dementia incidence, across genetic risk groups. In this analysis, follow-up time began at age at visit 4. Third, we examined the association of less severe hypertension with a less restrictive BP cutoff (SBP ≥ 120 mm Hg, DBP ≥ 90 mm Hg, or use of blood pressure medication in the past 2 weeks) and dementia incidence across genetic risk groups. Fourth, we examined the association of antihypertensive use only and dementia incidence across genetic risk groups. Finally, because early neurodegeneration may lead to lower blood pressure,²⁵ we introduced a 5-year lag by changing the time origin to age 60 years.

Standard Protocol Approvals, Registrations, and Participant Consents

The study was approved by each institution's Institutional Review Board. All participants provided written informed consent.

Data Availability

ARIC data are available through BioLINCC and dbGaP.

Results

Participant Characteristics

At visit 1 (total $n = 11,597$), participants identified as 23% Black and 77% White, with mean ages of 53.5 years among Black participants and 54.3 years among White participants. At baseline, both White and Black participants with hypertension were older, had lower education levels, and had more vascular risk factors than their nonhypertensive counterparts (Table 1). White participants had a median follow-up time of 26.6 years, a total of 2,106 incident cases of dementia, and 4,026 deaths without dementia. Black participants had a median follow-up time of 23.8 years, a total of 759 incident cases of dementia, and 1,227 deaths without dementia.

Evaluation of AD Genetic Risk Group and Incident Dementia

After adjusting for model 1 covariates, each standard deviation increase in AD PRS was associated with a higher risk of dementia for White (hazard ratio [HR] 1.40; 95% CI 1.34–1.46) and Black participants (HR 1.29; 95% CI 1.20–1.38) (Table 2). Similarly, participants in the high genetic risk category were at higher risk of dementia compared with the low genetic risk category for White (HR 1.95; 95% CI 1.76–2.17) and Black participants (HR 1.67; 95% CI 1.42–1.98); White, but not Black, individuals in the medium genetic risk category also had

a statistically significantly increased risk of dementia. After adjusting for model 2 covariates (eTable 3), and after removing variants in the *APOE* region from the PRS derivation (eTable 4), PRS standard deviations and tertiles were still statistically significantly associated with dementia in both race groups.

Evaluation of Hypertension and Incident Dementia by AD Genetic Risk Group

Among both White and Black participants, and among all AD genetic risk groups, there was a higher incidence rate of dementia for participants with hypertension compared with participants without hypertension (Figure 1). Among White participants, after adjusting for model 1 covariates, hypertension was significantly associated with a higher risk of dementia for individuals at low genetic risk of AD (HR 1.29; 95% CI 1.07–1.55), medium genetic risk of AD (HR 1.34; 95% CI 1.13–1.58), and high genetic risk of AD (HR 1.19; 95% CI 1.03–1.38) (Figure 2). Among Black participants, hypertension was significantly associated with a higher risk of dementia for individuals at high genetic risk of AD (HR 1.31; 95% CI 1.04–1.66). For Black participants at low (HR 1.10; 95% CI 0.86–1.40) and medium genetic risk of AD (HR 1.31; 95% CI 0.98–1.76), the effect size was similar between hypertension and incident dementia as it was for the other genetic risk and race groups, but these associations were not statistically significant. In both race groups, there were no significant differences in association of hypertension with dementia by AD genetic risk group. When this analysis was repeated after adjusting for model 2 covariates, the estimates were similar but slightly attenuated (eTable 5).

Adjusted Probability of Dementia and Dementia-Free Survival by Age 80 Years

The impact of hypertension on probability of dementia by age 80 years was similar across AD genetic risk groups and across race groups. Among White individuals, those at high genetic risk with midlife hypertension had the highest probability of developing dementia by age 80 (6.55%; 95% CI 5.1–8.1), with midlife hypertension associated with a 0.5%–1.0% increased risk across all genetic groups. Similarly, among Black individuals, the highest dementia probability was observed in the high genetic risk, hypertensive group (10.84%; 95% CI 7.5–14.4), with midlife hypertension linked to a 0.07%–1.7% higher risk of dementia in each genetic risk group (Figure 3A).

The likelihood of living to age 80 years without developing dementia was highest for individuals at low genetic risk of AD and without midlife hypertension: 71.77% (95% CI 69.6–74.4) for White individuals and 63.38% (95% CI 58.4–69.0) for Black individuals. In all genetic risk categories, midlife hypertension was associated with an 8%–10% lower probability of surviving to age 80 without dementia for White individuals and a 6%–13% reduction for Black individuals (Figure 3B).

We also estimated the probability of dementia, death without dementia, and dementia-free survival by ages 70 and 90 years (eTable 6). By age 70, the individuals with midlife

Table 1 White and Black Participant Characteristics by Visit 1 Hypertension Status

	White participants (n = 8,931)			Black participants (n = 2,666)		
	No hypertension (n = 6,533)	Hypertension (n = 2,398)	p Value	No hypertension (n = 1,179)	Hypertension (n = 1,487)	p Value
Age at visit 1, y, mean (SD)	53.6 (5.61)	56.2 (5.47)	<0.001	52.2 (5.63)	54.5 (5.68)	<0.001
Decade of birth, n (%)						
1920–29	1,547 (23.7)	979 (40.8)	<0.001	196 (16.6)	409 (27.5)	<0.001
1930–39	3,435 (52.6)	1,134 (47.3)		584 (49.5)	782 (52.6)	
1940+	1,551 (23.7)	285 (11.9)		399 (33.8)	296 (19.9)	
Female sex, n (%)	3,515 (53.8)	1,219 (50.8)	0.014	718 (60.9)	947 (63.7)	0.151
Education level, n (%)						
<HS	916 (14.0)	460 (19.2)	<0.001	401 (34.0)	674 (45.3)	<0.001
HS or GED or vocational school	2,936 (44.9)	1,123 (46.8)		345 (29.3)	399 (26.8)	
>HS	2,681 (41.0)	815 (34.0)		433 (36.7)	414 (27.8)	
BMI, kg/m ² , mean (SD)	26.3 (4.45)	28.8 (5.37)	<0.001	28.5 (5.52)	30.7 (6.28)	<0.001
Total cholesterol, mg/dL, mean (SD)	213 (40.0)	221 (42.3)	<0.001	212 (42.7)	219 (46.0)	<0.001
Current smoker, n (%)	1,719 (26.3)	484 (20.2)	<0.001	361 (30.6)	417 (28.0)	0.158
Diabetes, n (%)	377 (5.8)	394 (16.4)	<0.001	140 (11.9)	371 (24.9)	<0.001
AD PRS categories, n (%)						
Low risk	2,172 (33.2)	797 (33.2)	0.94	486 (41.2)	639 (43.0)	0.365
Medium risk	2,173 (33.3)	806 (33.6)		315 (26.7)	362 (24.3)	
High risk	2,188 (33.5)	795 (33.2)		378 (32.1)	486 (32.7)	

Abbreviations: AD = Alzheimer disease; BMI = body mass index; GED = general educational development; HS = high school; PRS = Polygenic Risk Score. *p* Values for continuous variables: *t* test. *p* Values for categorical or binary variables: χ^2 test.

hypertension had a higher CIF of dementia compared with individuals without hypertension across all genetic risk groups, but the difference was negligible. By age 90 years, individuals with hypertension were more likely to die, and the influence of mortality differences dominated the CIF of dementia estimates.

Sensitivity Analyses

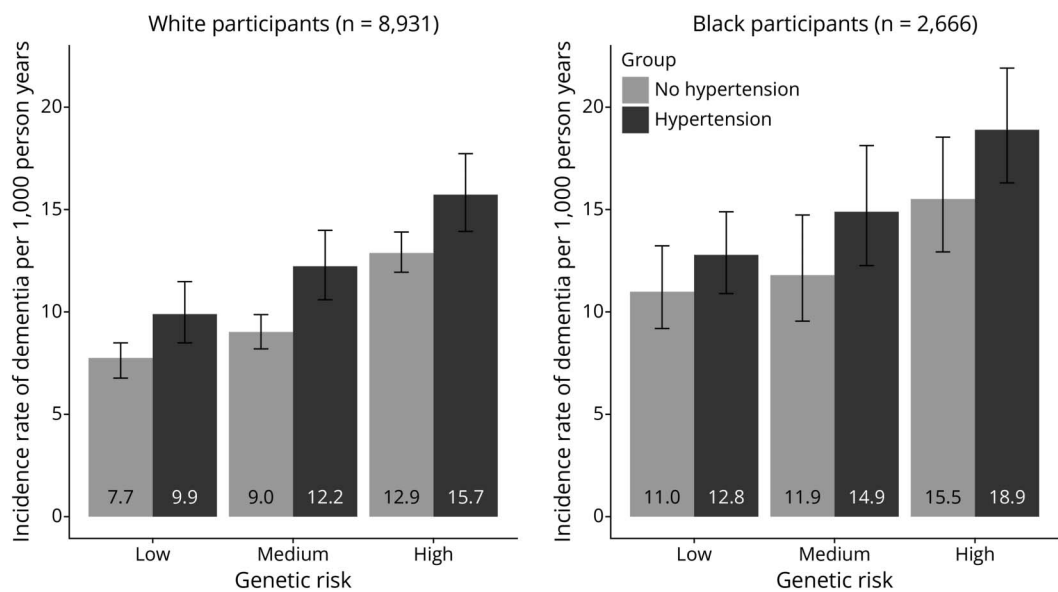
When hypertension was defined by measured high blood pressure at visit 1 alone, it was not significantly associated with dementia in any genetic risk group in either race group (eTable 7). Measured high blood pressure at any of the first 4 visits was only significantly associated with dementia in the

Table 2 Association Between AD Polygenic Risk Score and Dementia Incidence

	Model 1: Adjusted HR (95% CI)	
	White participants (n = 8,931)	Black participants (n = 2,666)
AD PRS, per SD	1.40 (1.34–1.46)	1.28 (1.20–1.37)
AD PRS tertiles		
Low risk	Ref	Ref
Medium risk	1.28 (1.14–1.43)	1.17 (0.97–1.41)
High risk	1.95 (1.76–2.17)	1.68 (1.42–1.98)

Abbreviations: AD = Alzheimer disease; HR = hazard ratio; PRS = Polygenic Risk Score. Origin is age 55 years, age as time scale. Model 1 covariates: sex, decade of birth, and education.

Figure 1 Incidence Rates (95% CIs) of Dementia by AD Genetic Risk Group and Hypertension Status



AD = Alzheimer disease.

high and medium genetic risk groups in White participants (eTable 8), but the effect sizes were similar to the effect of high blood pressure measured only at visit 1. Less severe hypertension was associated with dementia in White participants, but not in Black participants (eTable 9).

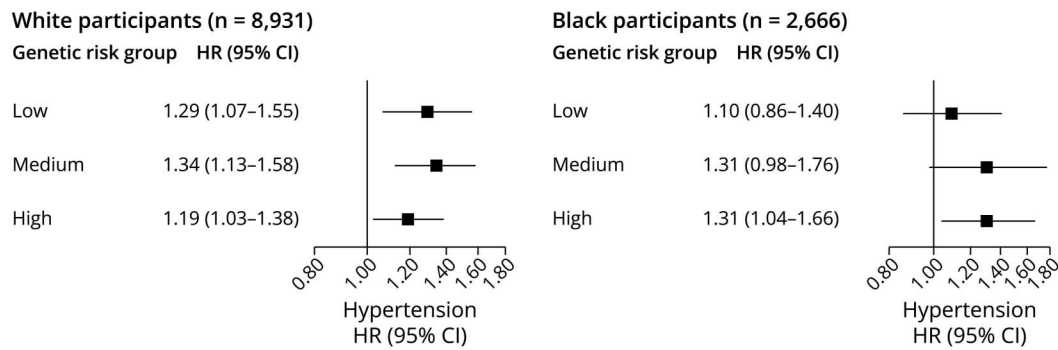
On the other hand, the effect of antihypertensive use alone (without considering measured blood pressure) on incident dementia was slightly stronger than the main results for individuals at low AD genetic risk (White HR 1.33; 95% CI 1.08–1.62, Black HR 1.15; 95% CI 0.90–1.45), medium AD genetic risk (White HR 1.46; 95% CI 1.21–1.75, Black HR 1.21; 95% CI 0.90–1.63), and high AD genetic risk (White HR 1.23; 95% CI 1.05–1.45, Black HR 1.25; 95% CI 0.99–1.58) (eTable 10).

Finally, using a 5-year lag to assess for the possibility of reverse causation, the estimates were consistent with the main analysis (eTable 11).

Discussion

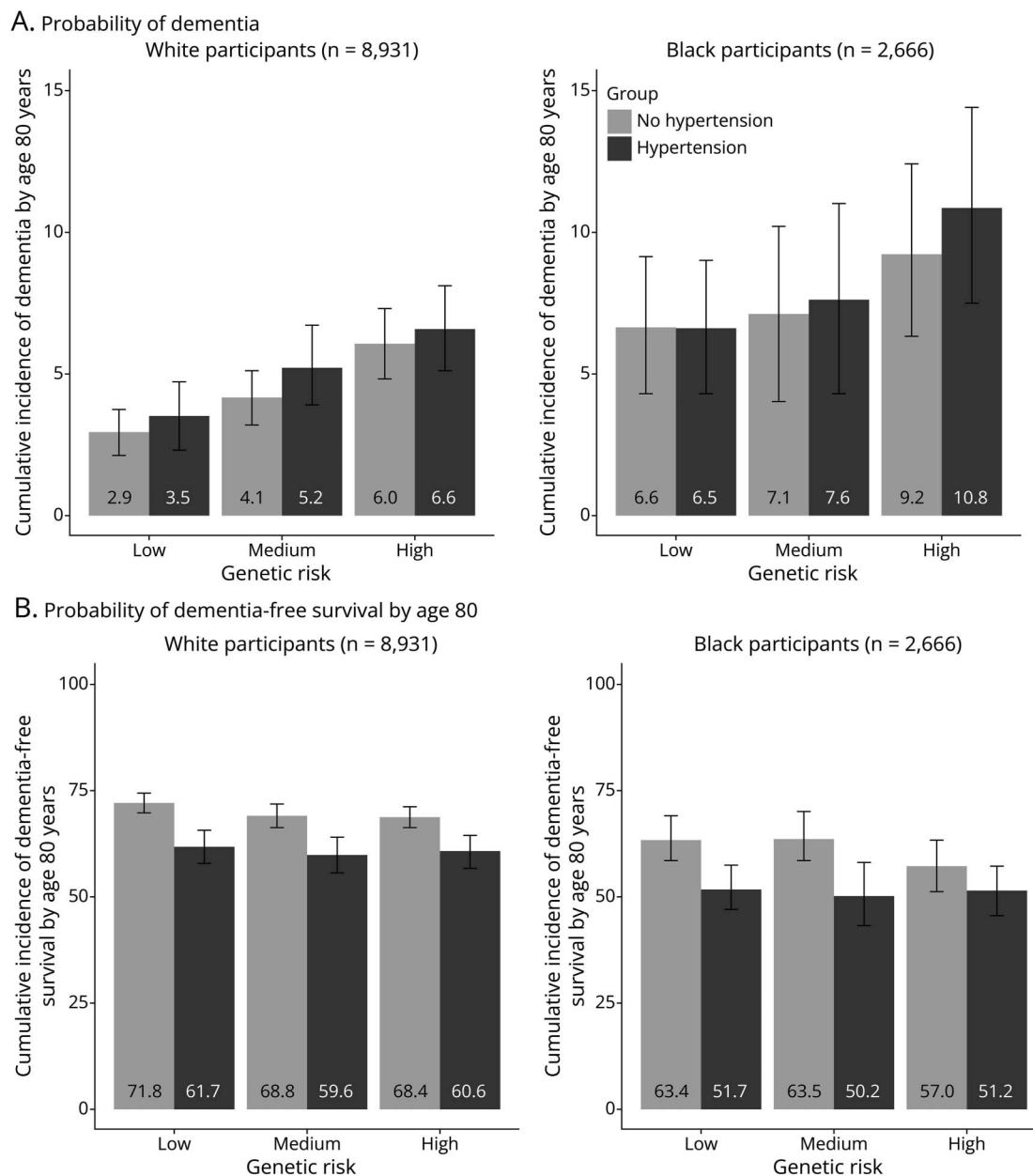
In a large community-based cohort of White and Black adults older than 55 years, we evaluated whether genetic risk for AD modified the association between hypertension and syndromic dementia. Midlife hypertension was significantly associated with a higher incidence of dementia within all AD genetic risk strata for White participants. The associations for Black participants were similar in effect size, but not always significant, likely due to smaller sample size.

Figure 2 Association Between Hypertension and Dementia Incidence Stratified by AD Genetic Risk Group



Six cox proportional hazards models, stratified by AD genetic risk group (AD PRS tertile) and race. Origin is age 55 years, age as time scale. Model 1 covariates: sex, decade of birth, and education. *p* Values for the interaction between genetic risk strata (medium vs low and high vs low) and hypertension were 0.790 and 0.505 in White participants, and 0.271 and 0.223 in Black participants respectively. AD = Alzheimer disease; HR = hazard ratio; PRS = polygenic risk score.

Figure 3 Estimated Probability of Dementia and Dementia Free Survival by Age 80 by AD Genetic Risk Groups and Midlife Hypertension



Mean centered covariates: sex, decade of birth, education. AD = Alzheimer disease.

Inconsistent results have been published on the associations of hypertension on dementia risk across AD genetic risk groups. In 2 previous ARIC studies, the association of hypertension with incident dementia was similar between *APOE*ε4 carriers and noncarriers.^{5,22} However, more recent work in ARIC found that there was a larger vascular contribution to dementia incidence among non-*APOE*ε4 carriers.²⁶ Unlike these studies, we used a PRS to assign genetic risk group, capturing genetic information beyond *APOE*ε4 carrier status (eTable 4). Our findings are in line with a previous study by Littlejohns et al.,²⁷ who examined the association of hypertension with dementia risk across dementia PRS

subgroups in the UK Biobank. They found that the impact of midlife hypertension on 12-year incident dementia did not differ by PRS subgroup. Notably, the association of hypertension with dementia incidence in the UK Biobank was similar in strength to that which we observed here. However, our study adds new information by including Black participants, using 35 years of follow-up, and a more rigorous ascertainment of dementia cases.

Although patients at high genetic risk for dementia may perceive a lack of control over their own dementia outcomes,²⁸ this study supports that they may still be able to

prevent or delay dementia by controlling their blood pressure in midlife in coordination with their healthcare providers. In addition, the absolute risk estimates presented here may be helpful for individuals and their clinicians to make informed decisions about changes to their lifestyle. Furthermore, the results of this study suggests that Black and White individuals in all genetic risk groups can reduce their probability of dementia by age 80 years by about 1% and increase their probability of surviving to age 80 years without dementia by about 10% by preventing hypertension in midlife. Importantly, the potential reduction in dementia alone is relatively small because many individuals with midlife hypertension die of other hypertension-related complications before developing dementia, which is why consideration of dementia-free survival is a more compelling statistic, and reflects a desire for improved quality of life in older age, and not merely additional years of life.

Our study has several limitations. First, we characterized midlife hypertension at 1 time point in our primary analyses, which does not account for potential cumulative effects and might leave residual confounding. However, associations of midlife hypertension with dementia incidence were consistent with the main results when the first 4 study visits were used to define hypertension (eTable 8). Second, we defined genetic risk groups based on tertile of an AD PRS, which does not capture all genetic variation linked to AD. Our race-specific PRS was based on 2 GWAS which did not capture rare variation or interacting genetic effects. In addition, the sample size of the GWAS among African ancestry individuals was smaller than the sample size of the GWAS among European ancestry individuals, and therefore, it detected far fewer significant loci. This disparity reflects systemic bias and longstanding historical underrepresentation of non-European ancestry individuals in genetic research. Genetic risk stratification could be improved, especially for Black participants, with the availability of larger, multiethnic GWAS in the future. Third, while the AD PRS was intended to capture genetic risk for AD, it may also capture genetic risk for other types of dementias due to lack of precision in phenotyping; similarly, we evaluated all-cause dementia as our outcome but it is possible that associations with the PRS (but perhaps not with hypertension) may have been stronger were these classified as dementia due to AD. Fourth, while we found potential differences in effect of midlife hypertension across race groups, we could not directly compare them because we used different AD PRS to define race specific AD genetic risk groups. Finally, our study was restricted to individuals who identified as Black or White race. Additional studies are needed to replicate these results in other groups.

In conclusion, midlife hypertension was associated with elevated risk of dementia irrespective of genetic risk for AD. Our findings indicate vascular risk might impact the clinical expression of dementia independently of Alzheimer's processes. Blood pressure control as early as midlife could significantly reduce risk of dementia.

Acknowledgment

The authors thank the staff and participants of the ARIC study for their important contributions.

Author Contributions

V.N. Morrill: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. J.R. Pike: study concept or design; analysis or interpretation of data. J. Hu: analysis or interpretation of data. M. Fornage: drafting/revision of the manuscript for content, including medical writing for content; study concept or design. A. Surapaneni: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K.A. Walker: drafting/revision of the manuscript for content, including medical writing for content. D.S. Knopman: drafting/revision of the manuscript for content, including medical writing for content. T.H. Mosley: drafting/revision of the manuscript for content, including medical writing for content. J. Coresh: drafting/revision of the manuscript for content, including medical writing for content. A.L.C. Schneider: drafting/revision of the manuscript for content, including medical writing for content. J.R. Smith: drafting/revision of the manuscript for content, including medical writing for content. R.F. Gottesman: drafting/revision of the manuscript for content, including medical writing for content; major role in the acquisition of data; study concept or design; analysis or interpretation of data.

Study Funding

The Atherosclerosis Risk in Communities Study is carried out as a collaborative study supported by National Heart, Lung, and Blood Institute contracts (75N92022D00001, 75N92022D00002, 75N92022D00003, 75N92022D00004, 75N92022D00005). The ARIC Neurocognitive Study is supported by U01HL096812, U01HL096814, U01HL096899, U01HL096902, and U01HL096917 from the NIH (NHLBI, National Institute of Neurological Disorders and Stroke, NIA and NIDCD). Funding was also supported by R01HL087641 and R01HL086694; National Human Genome Research Institute contract U01HG004402; and NIH contract HHSN268200625226C. Infrastructure was partly supported by Grant Number UL1RR025005, a component of the NIH and NIH Roadmap for Medical Research. J. Hu was supported by the National Natural Science Foundation of China (Grant Number 72504161). V.N. Morrill and R.F. Gottesman are supported by the National Institute of Neurological Disorders and Stroke intramural research program and K.A. Walker is supported by the NIA intramural research program. This research was supported in part by the Intramural Research Program of the NIH. The contributions of the NIH author(s) were made as part of their official duties as NIH federal employees, are in compliance with agency policy requirements, and are considered Works of the United States Government. However, the findings and conclusions presented in this paper are those of the author(s) and do not

necessarily reflect the views of the NIH or the U.S. Department of Health and Human Services. It is subject to the NIH Public Access Policy. Through acceptance of this federal funding, NIH has been given a right to make this manuscript publicly available in PubMed Central upon the Official Date of Publication, as defined by NIH.

Disclosure

D. Knopman serves on a data safety monitoring board for the Dominantly Inherited Alzheimer Network Treatment Unit study sponsored by Washington University St. Louis, and for the SMART-HS clinical trial (University of Kentucky). He was an investigator in Alzheimer clinical trials sponsored by Biogen, Lilly Pharmaceuticals and the University of Southern California, all of which have ended, and is currently an investigator in a trial in frontotemporal degeneration with Alector. He has served as a consultant for Roche, AriBio, Biovie and Alzeca Biosciences but receives no personal compensation. He receives funding from the NIH. All other authors report no disclosures relevant to the manuscript. Go to [Neurology.org/N](https://www.neurology.org/N) for full disclosures.

Publication History

Received by *Neurology*® November 21, 2025. Accepted in final form May 1, 2026. Submitted and externally peer reviewed. The handling editor was Associate Editor Linda Hershey, MD, PhD, FAAN.

References

- Manly JJ, Jones RN, Langa KM, et al. Estimating the prevalence of dementia and mild cognitive impairment in the US: the 2016 health and retirement study harmonized cognitive assessment protocol project. *JAMA Neurol.* 2022;79(12):1242-1249. doi:10.1001/jamaneurol.2022.3543
- Livingston G, Huntley J, Liu KY, et al. Dementia prevention, intervention, and care: 2024 report of the Lancet standing Commission. *Lancet.* 2024;404(10452):572-628. doi:10.1016/s0140-6736(24)01296-0
- Gottesman RF, Egle M, Groechel RC, Mughal A. Blood pressure and the brain: the conundrum of hypertension and dementia. *Cardiovasc Res.* 2025;120(18):2360-2372. doi:10.1093/cvr/cvaf010
- Osthega Y, Fryar C, Nwankwo T, Nguyen D. Hypertension prevalence among adults aged 18 and over: United States, 2017-2018. *NCHS Data Brief.* 2020(364):1-8.
- Walker KA, Sharrett AR, Wu A, et al. Association of midlife to late-life blood pressure patterns with incident dementia. *JAMA.* 2019;322(6):535-545. doi:10.1001/jama.2019.10575
- Walker KA, Power MC, Gottesman RF. Defining the relationship between hypertension, cognitive decline, and dementia: a review. *Curr Hypertens Rep.* 2017;19(3):24. doi:10.1007/s11906-017-0724-3
- Gottesman RF, Schneider AL, Albert M, et al. Midlife hypertension and 20-year cognitive change: the Atherosclerosis Risk in Communities Neurocognitive Study. *JAMA Neurol.* 2014;71(10):1218-1227. doi:10.1001/jamaneurol.2014.1646
- Gatz M, Reynolds CA, Fratiglioni L, et al. Role of genes and environments for explaining Alzheimer disease. *Arch Gen Psychiatry.* 2006;63(2):168-174. doi:10.1001/archpsyc.63.2.168
- Saunders AM, Strittmatter WJ, Schmechel D, et al. Association of apolipoprotein E allele epsilon 4 with late-onset familial and sporadic Alzheimer's disease. *Neurology.* 1993;43(8):1467-1472. doi:10.1212/wnl.43.8.1467
- Bellenguez C, Küçükali F, Jansen IE, et al. New insights into the genetic etiology of Alzheimer's disease and related dementias. *Nat Genet.* 2022;54(4):412-436. doi:10.1038/s41588-022-01024-z
- Sherva R, Zhang R, Sahelijo N, et al. African ancestry GWAS of dementia in a large military cohort identifies significant risk loci. *Mol Psychiatry.* 2023;28(3):1293-1302. doi:10.1038/s41380-022-01890-3
- Tin A, Bressler J, Simino J, et al. Genetic risk, midlife life's simple 7, and incident dementia in the Atherosclerosis Risk in Communities Study. *Neurology.* 2022;99(2):e154-e163. doi:10.1212/wnl.0000000000200520
- Yu C, Ryan J, Orchard SG, et al. Validation of newly derived polygenic risk scores for dementia in a prospective study of older individuals. *Alzheimers Dement.* 2023;19(12):5333-5342. doi:10.1002/alz.13113
- Lourida I, Hannon E, Littlejohns TJ, et al. Association of lifestyle and genetic risk with incidence of dementia. *JAMA.* 2019;322(5):430-437. doi:10.1001/jama.2019.9879
- Marden JR, Walter S, Tchetgen Tchetgen EJ, Kawachi I, Glymour MM. Validation of a polygenic risk score for dementia in black and white individuals. *Brain Behav.* 2014;4(5):687-697. doi:10.1002/brb.3.248
- Jung SH, Kim HR, Chun MY, et al. Transferability of Alzheimer disease polygenic risk score across populations and its association with Alzheimer disease-related phenotypes. *JAMA Netw Open.* 2022;5(12):e2247162. doi:10.1001/jamanetworkopen.2022.47162
- Wright JD, Folsom AR, Coresh J, et al. The ARIC (Atherosclerosis Risk In Communities) study: JACC focus seminar 3/8. *J Am Coll Cardiol.* 2021;77(23):2939-2959. doi:10.1016/j.jacc.2021.04.035
- Rhee EP, Surapaneni AL, Schlosser P, et al. A genome-wide association study identifies 41 loci associated with eicosanoid levels. *Commun Biol.* 2023;6(1):792. doi:10.1038/s42003-023-05159-5
- Purcell S, Neale B, Todd-Brown K, et al. PLINK: a tool set for whole-genome association and population-based linkage analyses. *Am J Hum Genet.* 2007;81(3):559-575. doi:10.1086/519795
- Knopman DS, Gottesman RF, Sharrett AR, et al. Mild cognitive impairment and dementia prevalence: the Atherosclerosis Risk in Communities Neurocognitive Study (ARIC-NCS). *Alzheimers Dement (Amst).* 2016;2:1-11. doi:10.1016/j.dadm.2015.12.002
- Knopman DS, Pike JR, Gottesman RF, et al. Patterns of cognitive domain abnormalities enhance discrimination of dementia risk prediction: the ARIC study. *Alzheimers Dement.* 2024;20(7):4559-4571. doi:10.1002/alz.13876
- Gottesman RF, Albert MS, Alonso A, et al. Associations between midlife vascular risk factors and 25-year incident dementia in the Atherosclerosis Risk in Communities (ARIC) cohort. *JAMA Neurol.* 2017;74(10):1246-1254. doi:10.1001/jamaneurol.2017.1658
- Lewis CM, Vassos E. Polygenic risk scores: from research tools to clinical instruments. *Genome Med.* 2020;12(1):44. doi:10.1186/s13073-020-00742-5
- Hu J, Pike JR, Lutsey PL, et al. Age of diabetes diagnosis and lifetime risk of dementia: the Atherosclerosis Risk in Communities (ARIC) study. *Diabetes Care.* 2024;47(9):1576-1583. doi:10.2337/dc24-0203
- Kitamura J, Nagai M, Ueno H, et al. The insular cortex, Alzheimer disease pathology, and their effects on blood pressure variability. *Alzheimer Dis Assoc Disord.* 2020;34(3):282-291. doi:10.1097/wad.0000000000000340
- Smith JR, Pike JR, Gottesman RF, et al. Contribution of modifiable midlife and late-life vascular risk factors to incident dementia. *JAMA Neurol.* 2025;82(7):644-654. doi:10.1001/jamaneurol.2025.1495
- Littlejohns TJ, Collier JA, Liu X, Clifton L, Tapela NM, Hunter DJ. Hypertension, a dementia polygenic risk score, APOE genotype, and incident dementia. *Alzheimers Dement.* 2023;19(2):467-476. doi:10.1002/alz.12680
- Dar-Nimrod I, Heine SJ. Genetic essentialism: on the deceptive determinism of DNA. *Psychol Bull.* 2011;137(5):800-818. doi:10.1037/a0021860